Module 1

# Data Science

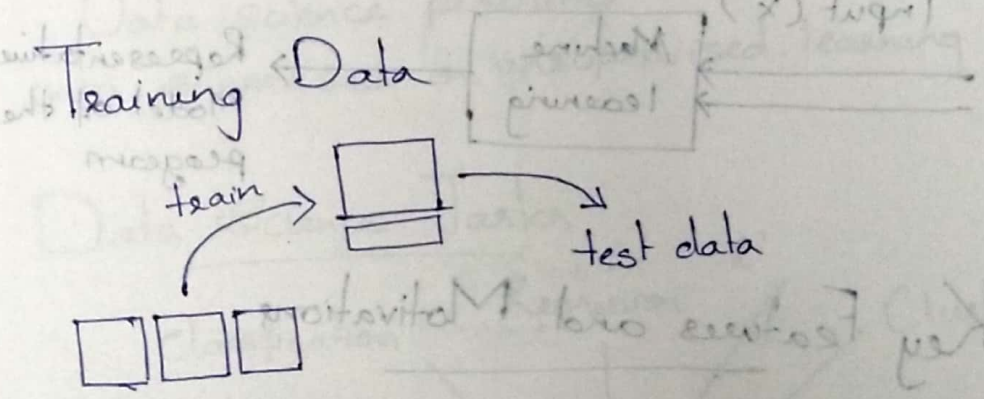
Massive data
Diversified data
Accumulated data
} are we need to manipulated

In data science data are used for

1. Prediction
2. to uncover hidden pattern
3. better decision

For analysing data we have to

1. Data gathering
2. Data preparation
3. Data cleaning
4. Data Transformation




## Artificial Intelligence

Linguistics
Vision
Language Synthesis
Robotics
Sensor

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# Machine learning

Support vector machines

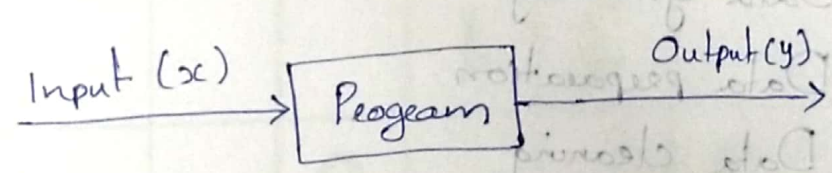
KNN.

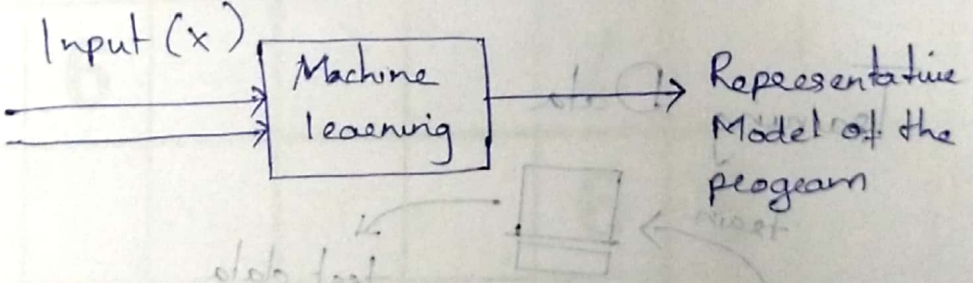
Decision trees

Bayesian learning

Deep

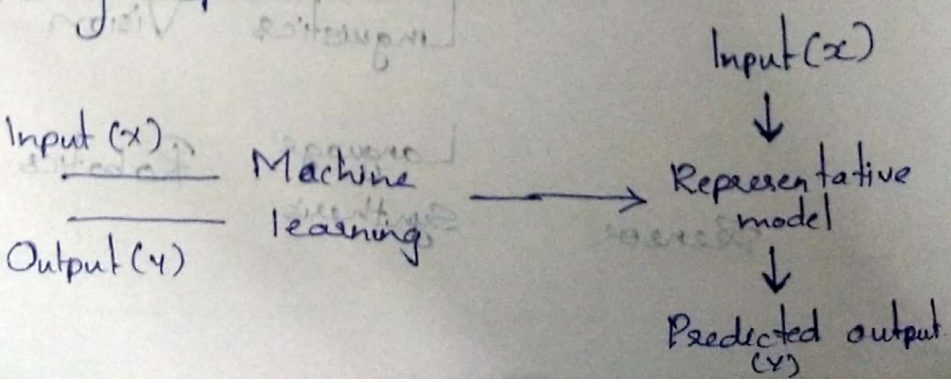
## Traditional peogeam and Machine Learning




## Key Features and Motivations

1. Extracting Mean
2. Building Representative Models




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3. Combination of Stats

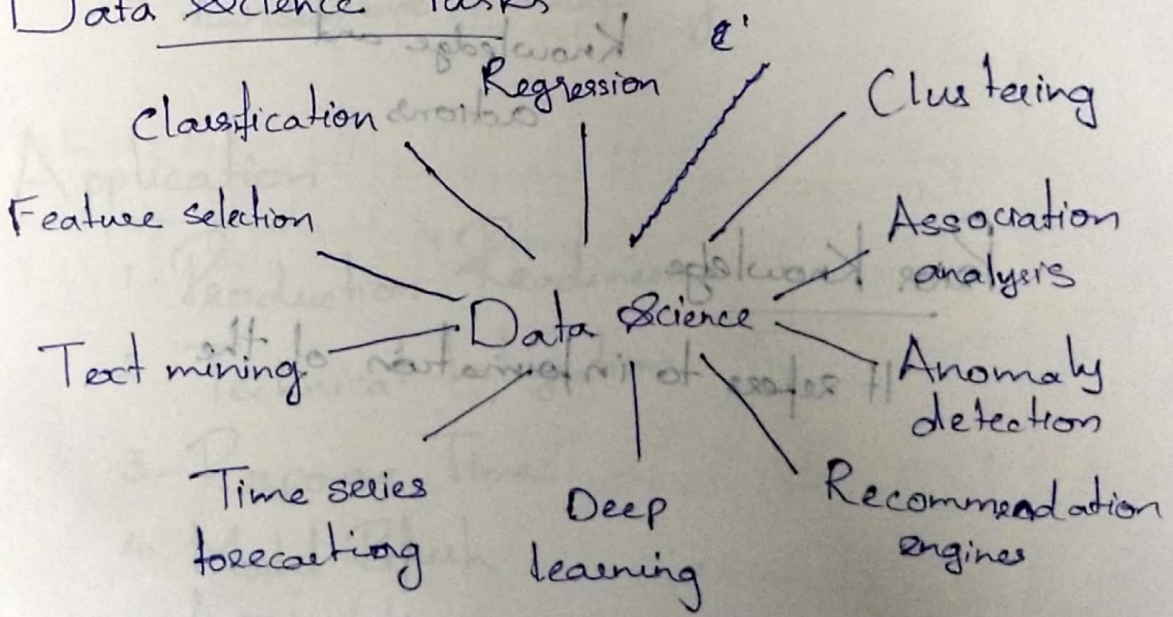
5. Associated Fields

1. Descriptive statistics
2. Exploratory visualization
3. Dimensional slicing
   - Slicing
   - Pivoting
   - Hypothesis testing
   - Data engineering
   - Business intelligence (BI)

## Data Science Classification

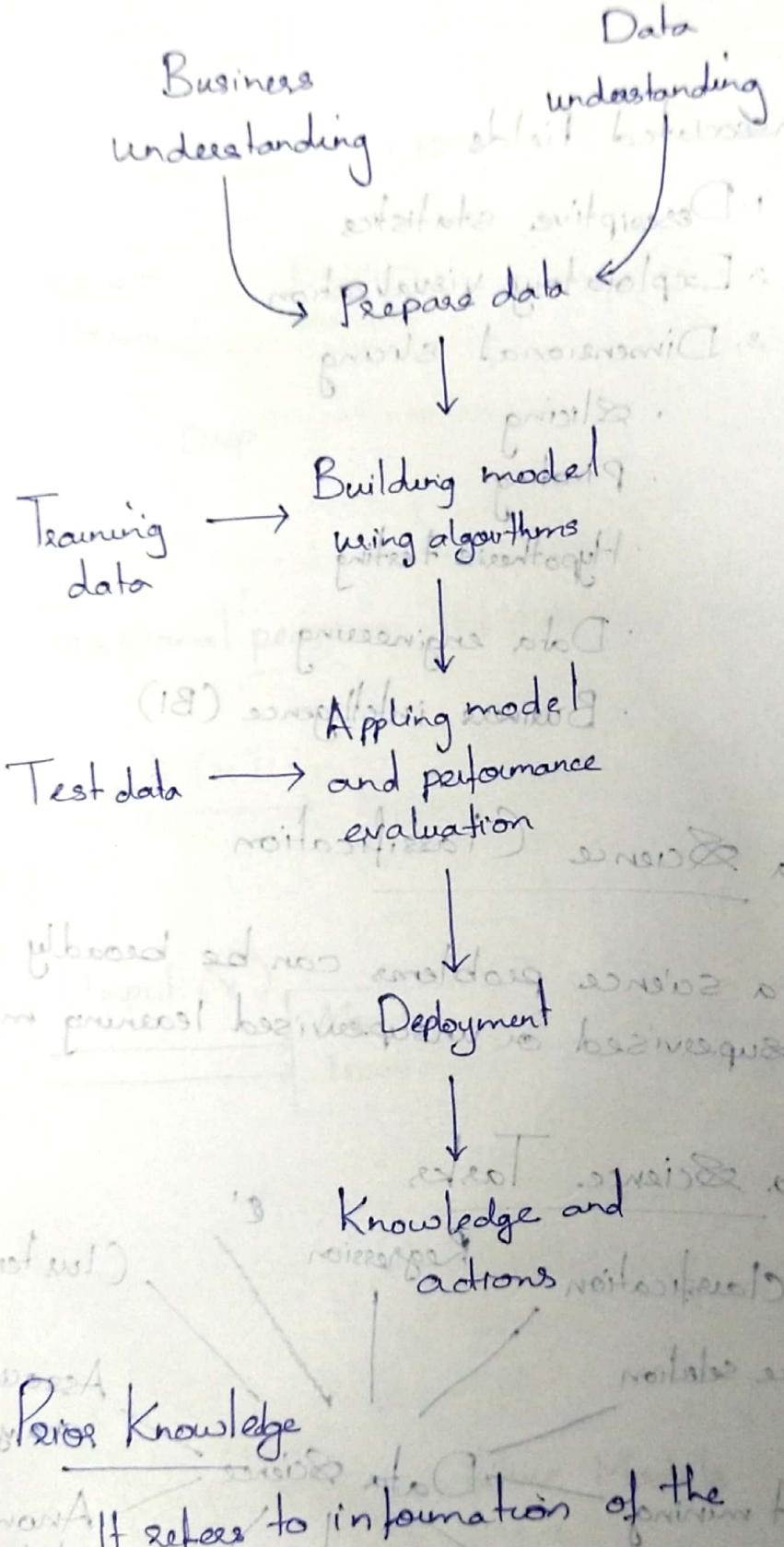
Data science problems can be broadly categorized into supervised or unsupervised learning models.

## Data Science Tasks




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# Data Science Process




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## Data Preparation

1. Data Exploration
2. Data Quality
3. Missing Values
4. Data Types and Conversion
5. Transformations
6. Outliers
7. Feature Selection
8. Data Sampling

## Modelling

1. Training dataset and Testing Datasets
2. Learning Algorithms
3. Evaluation of the Model
4. Ensemble Modeling

## Application

1. Production Readiness
2. Technical Integration
3. Response Time
4. Model Refresh
5. Assimilation

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# Knowledge

The data science process provide a framework to extract nontrivial information from data.

## Data Exploration

### Data Set

1. Descriptive Statistics
   2 types
   1. Univariate Exploration
   2. Measure of Spread

$$\text{Variance} : S^2 = \frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2$$

$x_i$ → Value; $\mu$ → mean of the Sample

Deviation :

$$\text{Standard deviation} : SD = \sqrt{\frac{\sum (x_i - \mu)}{N}}$$

2. Multivariate Exploration

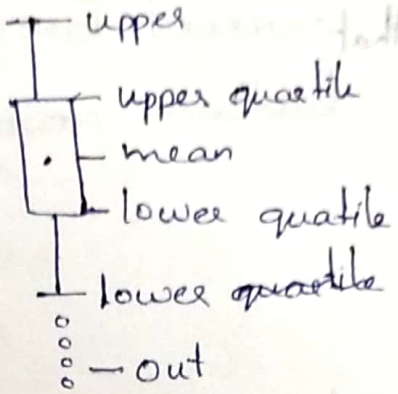
Correlation : It measures the statistical relationship between two attributes, particularly dependence of one attribute

## Data Visualization

1. Univariate
   1. Histogram
   2. Quartile

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Diagram shows that the quartile charts for all four attributes of the Iris dataset are plotted side by side. Petal length can be observed as having the broadest range and the sepal width has a narrow range, out of all of the four attributes




3. Distribution Chart

$$f(x) = \frac{1}{\sqrt{2\pi\sigma}} e^{(x-\mu)^2/2\sigma^2}$$

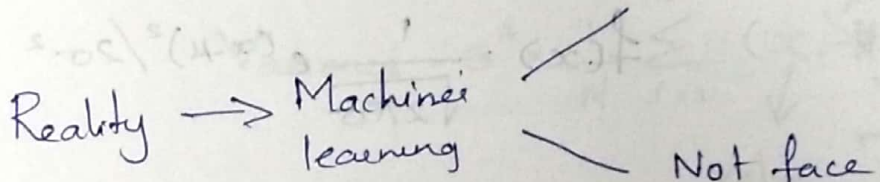
1. Scatterplot
   1. Iris setosa
   2. Iris versicolor
   3. Iris Virginica
2. Bubble Chart
3. Density Chart

# Module 2

## Machine Learning

The field of machine learning provides a set of algorithms that




## Machine Learning in Practice

1. Data collection
2. Data Exploration and preparation
3. Model training
4. Model evaluation
5. Model improvement

## Types of Input Data

1. Unit of observation
   Unit of analysis

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# Types of Machine Learning Algorithms

1. Predictive model

3 types of classification algorithms

1. Supervised Learnes
2. Regression method
3.
4.

1 Based on a survey conducted in an institution students are classified based on the 2 attribute of academic excellence and other activities. Given the following data identify the classification of a student with $x = 5$ and $y = 7$ using KNN algorithm (choose k as 3).

| | X (Academic Excellence) | Y (Other Activities) | Z (Classification) |
|---|---|---|---|
| Student 1 | 8 | 6 | Outstanding |
| Student 2 | 5 | 6 | Good |
| Student 3 | 7 | 3 | Good |
| Student 4 | 6 | 9 | Outstanding |
| Student 5 | 5 | 7 | ? |

$$\text{Distance}(P,Q) = \sqrt{(P_1-Q_1)^2 + (P_2-Q_2)^2 + \cdots + (P_k-Q_k)^2}$$

Let 'P' be student (5) with $x = 5$ and $y = 7$

'Q' be students who are already classified

$$\text{distance}(P,Q) = \sqrt{(P_1-Q_1)^2 + (P_2-Q_2)^2}$$

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dist (stud 5, stud 1) $= \sqrt{(5-8)^2 + (7-6)^2} = 3.162$

dist (stud 5, stud 2) $= \sqrt{(5-5)^2 + (7-6)^2} = 1$

dist (stud 5, stud 3) $= \sqrt{(5-7)^2 + (7-3)^2} = 4.472$

dist (stud 5, stud 4) $= \sqrt{(5-6)^2 + (7-9)^2} = 2.236$

| | X | Y | Z | K = 3 | |
|---|---|---|---|---|---|
| Stud 1 | 8 | 6 | 3.162 | Yes | Outstanding |
| stud 2 | 5 | 6 | 1 | Yes | Good |
| stud 3 | 7 | 3 | 4.472 | No | X |
| Stud 4 | 6 | 9 | 2.236 | Yes | Outstanding |

∴ Student 5 in an outstanding student

2 Perform KNN classification algorithm on following dataset and predict the class for X (P1 = 3 and P2 = 7) K = 3.

| P1 | P2 | Class |
|---|---|---|
| 7 | 7 | False |
| 7 | 4 | False |
| 3 | 4 | True |
| 1 | 4 | True |
| 3 | 7 | ? |

Distance $(P, Q) = \sqrt{(P_1 - Q_1)^2 + (P_2 - Q_2)^2 + \cdots + (P_n - Q_n)^2}$

Let 'P' be with P1 = 3 and P2 = 7

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dist (X, A) : $\sqrt{(3-7)^2 + (7-7)^2} = 4$

(X, B) : $\sqrt{(3-7)^2 + (7-4)^2} = 5$

(X, C) : $\sqrt{(3-3)^2 + (7-4)^2} = 3$

(X, D) : $\sqrt{(3-1)^2 + (7-4)^2} = 3.605$

| | $P_1$ | $P_2$ | Z | K = 3 | Class |
|---|---|---|---|---|---|
| A | 7 | 7 | 4 | Yes | False |
| B | 7 | 4 | 5 | No | False |
| C | 3 | 4 | 3 | Yes | True |
| D | 1 | 4 | 3.605 | Yes | True |

Majority True (C, D)

So, X is True

## Naive Bayes Algorithm

# Naive Bayes Problem / Classifier Problem

| Day | Outlook | Temperature | Humidity | Wind | PlayTennis |
|---|---|---|---|---|---|
| $D_1$ | Sunny | Hot | High | Weak | No |
| $D_2$ | Sunny | Hot | High | Strong | No |
| $D_3$ | Overcast | Hot | High | Weak | Yes |
| $D_4$ | Rain | Mild | High | Weak | Yes |
| $D_5$ | Rain | Cool | Normal | Weak | Yes |
| $D_6$ | ~~Overcast~~ Rain | Cool | Normal | Strong | No |
| $D_7$ | ~~Sunny~~ Overcast | Cool | Normal | Strong | Yes |
| $D_8$ | Sunny | Mild | High | Weak | No |
| $D_9$ | ~~Rain~~ Sunny | Cool | Normal | Weak | Yes |
| $D_{10}$ | Rain | Mild | Normal | Weak | Yes |
| $D_{11}$ | Sunny | Mild | Normal | Strong | Yes |
| $D_{12}$ | Overcast | Mild | High | Strong | Yes |
| $D_{13}$ | Overcast | Hot | Normal | Weak | Yes |
| $D_{14}$ | Rain | Mild | High | Strong | No |
| $D_{15}$ | Sunny | Cold | High | Strong | ? |

$P(\text{PlayTennis} = \text{yes}) = 9/14 = .64$

$P(\text{PlayTennis} = \text{no}) = 5/14 = .36$

| Outlook | Y | N |
|---|---|---|
| Sunny | 2/9 | 3/5 |
| Overcast | 4/9 | 0 |
| Rain | 3/9 | 2/5 |

| Temperature | Y | N |
|---|---|---|
| Hot | 2/9 | 2/5 |
| Mild | 4/9 | 2/5 |
| Cool | 3/9 | 1/5 |

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| Humidity | Y | N |
|---|---|---|
| High | 3/9 | 4/5 |
| Normal | 6/9 | 1/5 |

| Wind | Y | N |
|---|---|---|
| Weak | 6/9 | 2/5 |
| Strong | 3/9 | 3/9 |

(Outlook = Sunny, Temperature = Cool, Humidity = high, Wind = strong)

$$V_{NB} = \underset{v_j \in \{yes, no\}}{\operatorname{argmax}} P(v_j) \prod_i P(a_i / v_j)$$

$$= \underset{v_j \in \{yes, no\}}{\operatorname{argmax}} P(v_j)\, P(\text{Outlook} = \text{sunny} \mid v_i)\, P(\text{Temperature} = \text{cool} \mid v_j)$$

$$P(\text{Humidity} = \text{high} \mid v_j)\, P(\text{Wind} = \text{strong} \mid v_j)$$

$$V_{NB}(\text{Yes}) = P(\text{Yes})\, P(\text{Sunny} \mid \text{Yes})\, P(\text{Cool} \mid \text{Yes})\, P(\text{high} \mid \text{Yes})\, P(\text{strong} \mid \text{yes} = .0053$$

$$V_{NB}(\text{no}) = P(\text{no})\, P(\text{sunny} \mid \text{no})\, P(\text{cool} \mid \text{no})\, P(\text{high} \mid \text{no})\, P(\text{strong} \mid \text{no}) = .0206$$

$.64 \; (2 \div 9) \times (3 \div 9) \times (3 \div 9) \times (3 \div 9)$

$$P(\text{yes}) = \frac{.0053}{(.0053 + .0206)} = \frac{.0053}{0.0259} = 0.2046 = \underline{\underline{0.20}}$$

$$P(\text{no}) = \frac{.0206}{(.0206 + .0053)} = \frac{.0206}{0.0259} = \underline{\underline{0.81}}$$

The probability of the person playing tennis is classified as 'no' as the probability of outcome no for the given condition outlook = sun, humidity =

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high wind : strong ), we highest the probability of outcome Yes for the same condition.

## Example 2

Estimate conditional probabilities of each attributes {color, legs, height, smelly} for the species classes: {M, H} using the data given

Language

| No | Color | Legs | Height | Smelly | Species |
|---|---|---|---|---|---|
| 1 | White | 3 | Short | Yes | M |
| 2 | Green | 2 | Tall | No | M |
| 3 | Green | 3 | Short | Yes | M |
| 4 | White | 3 | Short | Yes | M |
| 5 | Green | 2 | Short | No | H |

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| | | | | | |
|---|---|---|---|---|---|
| 6 | White | 2 | Tall | No | H |
| 7 | White | 2 | Tall | No | H |
| 8 | White | 2 | Short | Yes | H |

New Instance

(Color = Green, legs = 2, Height = Tall, and Smelly = No)

$P(M) = \frac{4}{8} = 0.5$ $P(H) = \frac{4}{8} = 0.5$

| Color | M | H |
|---|---|---|
| White | 2/4 | 3/4 |
| Green | 2/4 | 1/4 |

| Legs | M | H |
|---|---|---|
| 2 | 1/4 | 4/4 |
| 3 | 3/4 | 0/4 |

| Height | M | H |
|---|---|---|
| Tall | 3/4 | 2/4 |
| Short | 1/4 | 2/4 |

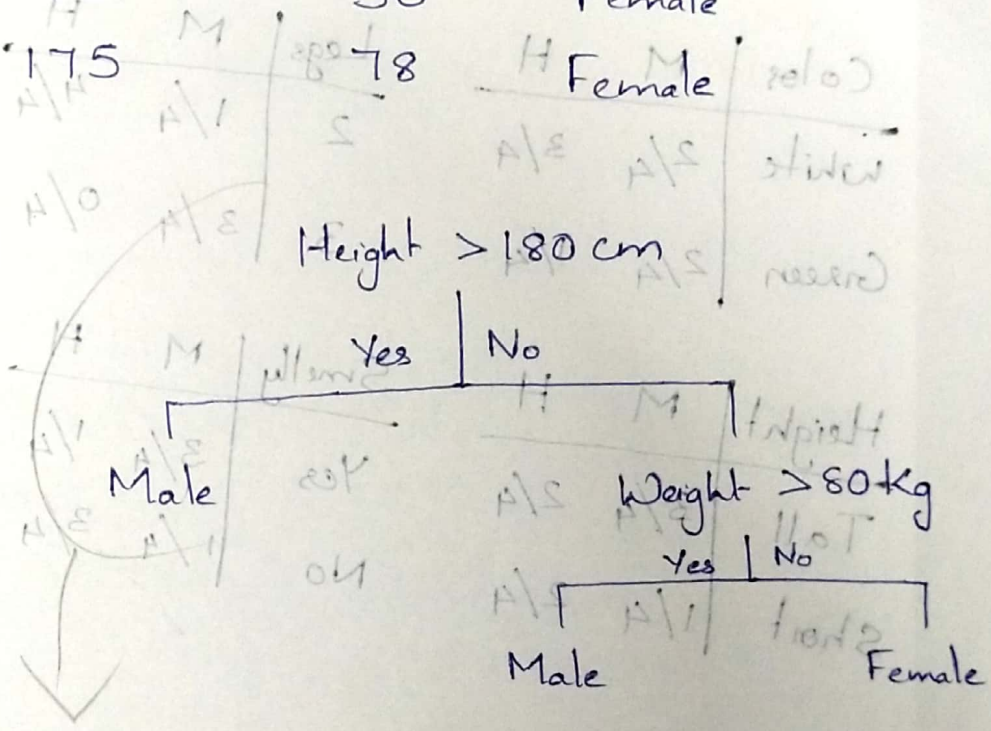
| Smelly | M | H |
|---|---|---|
| Yes | 3/4 | 1/4 |
| No | 1/4 | 3/4 |

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# Decision Tree

## Example

| Height (cms) | Weight (kg) | Class |
|---|---|---|
| 170 | 75 | Female |
| 170 | 90 | Male |
| 185 | 85 | Male |
| 155 | 50 | Female |
| 175 | 78 | Female |




| >180 | | | <180 | | |
|---|---|---|---|---|---|
| H | W | C | H | W | C |
| 180 | 85 | M | 170 | 75 | F |
| | | | 170 | 90 | M |
| | | | 155 | 50 | F |
| | | | 175 | 78 | F |

| >80kg | | | <80kg | | |
|---|---|---|---|---|---|
| H | W | C | H | W | C |
| 170 | 90 | M | 170 | 75 | F |
| | | | 155 | 50 | F |
| | | | 175 | 78 | F |

Entropy (S)

$$\sum_{i=1}^{c} -P_i (\log_2) P$$

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Entropy (S) $= \sum_{i=1}^{c} -P_i (\log_2) P_i$

↓

S Subset

c - No of class label to the subset S

$P_i$ - Propotion of ith class (how many samples belong to class i)

We need to sum up the value for each subset

Total Samples = 4

F

$-\frac{3}{4}(\log_2)\frac{3}{4} \; -\frac{1}{4}(\log_2)\frac{1}{4}$

$-\frac{3}{4} \times (-0.415) - \frac{1}{4} \times (-2)$

$= 0.81125$

| Instance | Classification | $a_1$ | $a_2$ |
|---|---|---|---|
| 1 | + | T | T |
| 2 | + | T | T |
| 3 | − | T | F |
| 4 | + | F | F |
| 5 | − | F | T |
| 6 | − | F | T |

a) Find the entropy of this collection of training eg wrt the fo function classification

b) Calculate the info gain of $a_2$ relative to the

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~~relative~~ to training example

$$\text{Entropy}(S) = \sum_{i=1}^{c} -P_i (\log_2) P_i$$

$$= -\frac{3}{6} (\log_2) \frac{3}{6}$$

$= 0.5 \rightarrow$ for +

$= \frac{-3}{6} (\log_2) \frac{3}{6} = 0.5 \rightarrow$ for -

$0.5 + 0.5 = 1.0$

$$\text{for +} = \frac{-4}{6} (\log_2) \frac{4}{6} =$$

$$\text{for -} = \frac{-2}{6} (\log_2) \frac{2}{6} =$$

$= 0.90$

Information gain (f) = $S_1 - S_2$

Entropy $(S_1)$ - Entropy $(S_2)$

$1 - 0.90 = 0.1$

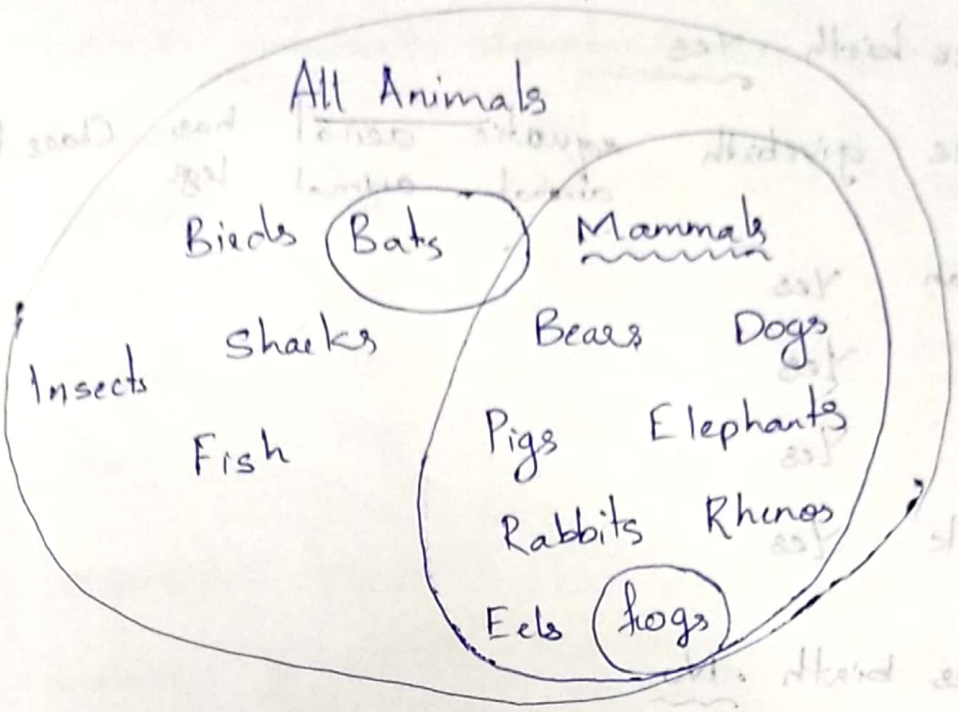
## Pruning the decision tree

1. Pre-pruning - early pruning
2. Post pruning

## Classification Rule

1. Separate and conquer

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# Classification Trees – Concept of Decision Tree

Example:

| Name | Features | | | | Class label |
|---|---|---|---|---|---|
| | gives birth | aquatic animal | aerial animal | has legs | |
| human | Yes | no | no | Yes | mammal |
| Python | no | no | no | no | reptile |
| Salmon | no | yes | no | no | fish |
| frog | no | semi | no | yes | amphibian |
| bat | Yes | no | yes | yes | bird |
| Pigeon | no | no | yes | yes | bird |
| cat | Yes | no | no | Yes | mammal |
| shark | yes | Yes | no | no | fish |
| turtle | no | semi | no | Yes | amphibian |
| Salamander | no | semi | no | Yes | amphibian |

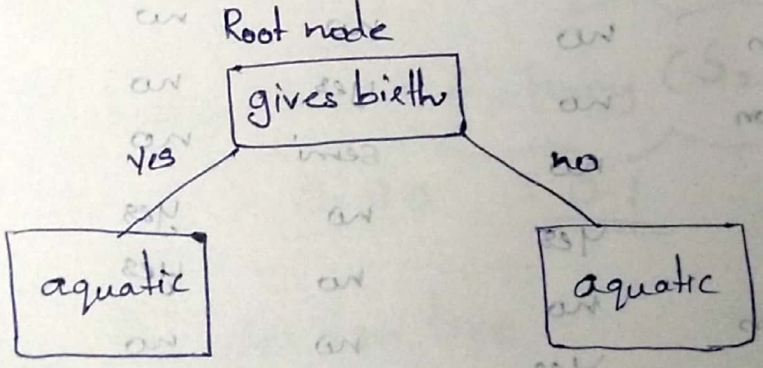
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gives birth : Yes

| name | givebirth | aquatic animal | aerial animal | has legs | Class label |
|---|---|---|---|---|---|
| human | Yes | | | | |
| bat | Yes | | | | |
| cat | Yes | | | | |
| Shark | Yes | | | | |

gives birth : No

| name | givebirth |
|---|---|
| Python | no |
| Salmon | no |
| frog | no |
| Pigeon | no |
| turtle | no |
| Salamander | no |




Gives birth : aquatic animal : Yes

| name | givebirth | aquatic |
|---|---|---|
| Shark | Yes | Yes |

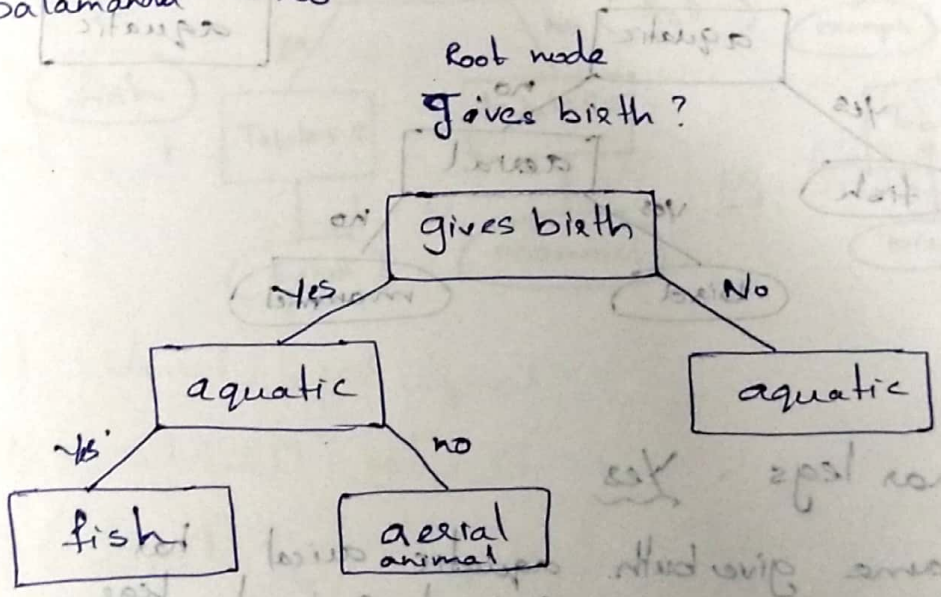
aquatic animal : No

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| name | give birth | aquatic animal |
|---|---|---|
| human | Yes | no |
| Python | no | no |
| bat | Yes | no |
| Pigion | no | no |
| cat | no | no |

aquatic animal : Semi

| name | give birth | aquatic animal |
|---|---|---|
| frog | Yes no | semi |
| turtle | no | semi |
| Salamander | no | semi |



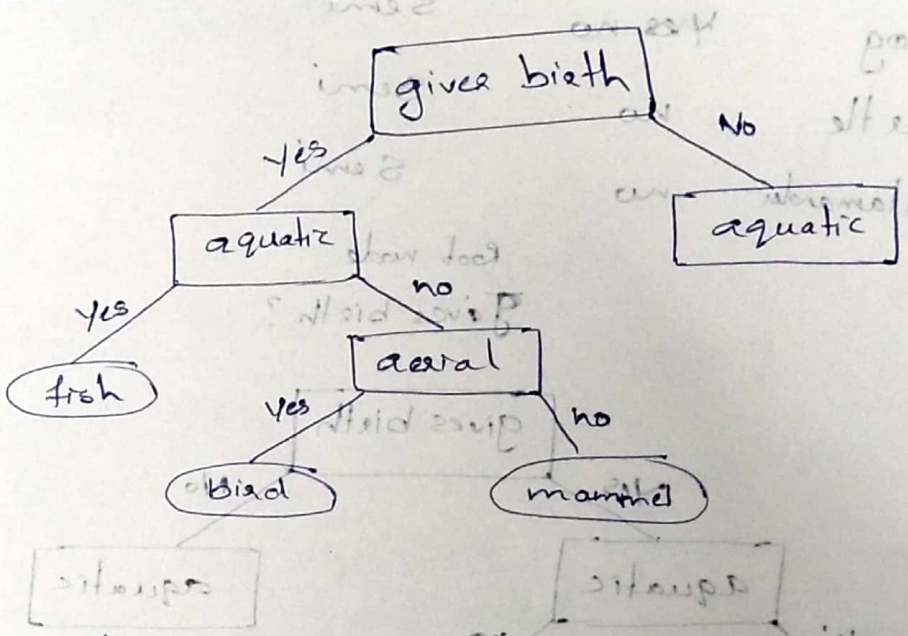

aerial animal : Yes

| name | give birth | aquatic animal | aerial animal |
|---|---|---|---|
| bat | Yes | no | yes |
| pegion | no | no | Yes |

aerial animal - No

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| name | gives birth | aquatic animal | aerial animal |
|---|---|---|---|
| human | Yes | no | no |
| python | no | no | no |
| frog | no | semi | no |
| Salamada | no | semi | no |
| cat | Yes | no | no |
| Shark | Yes | Yes | no |
| turtle | no | semi | no |
| Salamande | no | semi | no |




has legs : Yes

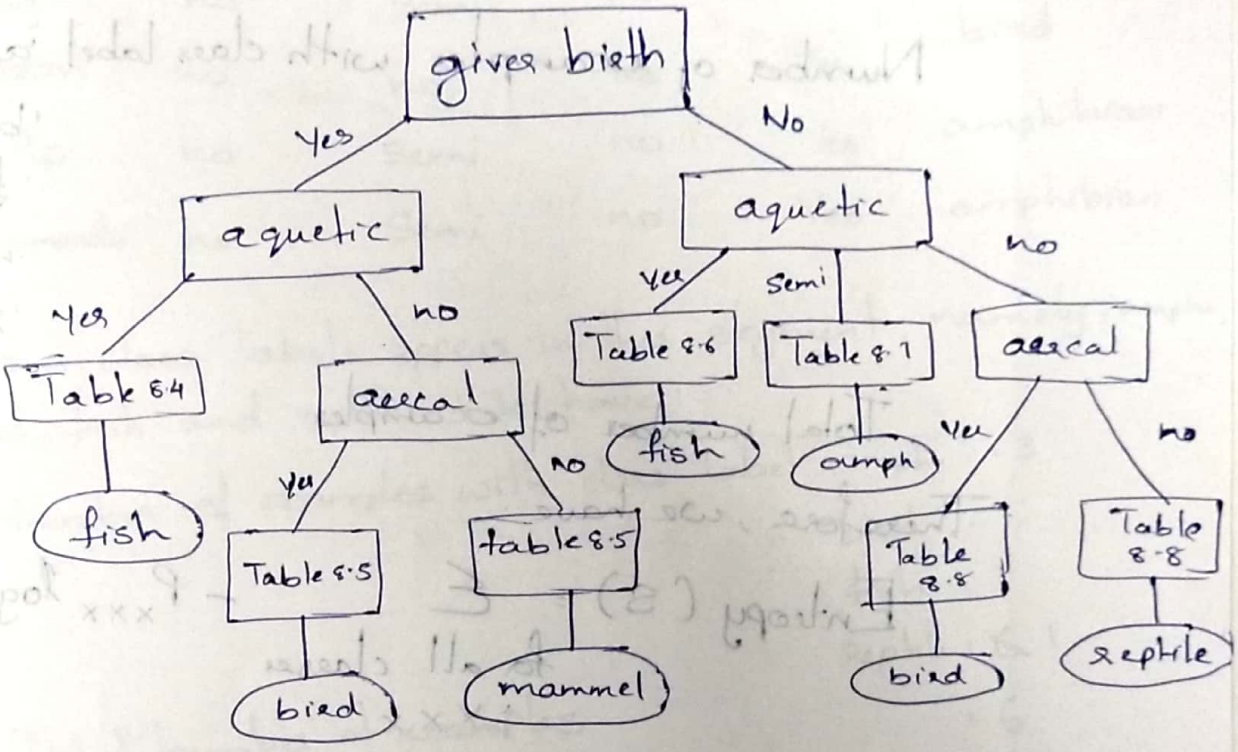
| name | give birth | aquatic animal | aerial animal | has legs |
|---|---|---|---|---|
| human | | | | Yes |
| frog | | | | Yes |
| bat | | | | Yes |
| Pigon | | | | Yes |
| cat | | | | Yes |
| turtle | | | | Yes |
| Salamande | | | | Yes |

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has legs . No

| name | give birth | aquatic animal | aercal animal | has legs |
|---|---|---|---|---|
| human | | | | no |
| python | | | | no |
| shark | | | | no |

Root node




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# Choosing the Best Split

## 1 Entropy of data

Let S be the data. The class labels are 'amphi', 'bird', 'fish', 'mammal' and 'reptile'. In S we have the following numbers.

Number of examples with class label 'amphi' : 3
'bird' : 2
'fish' : 2
'mammal' : 2
'reptile' : 1
10

Total number of examples

Therefore, we have:

$$\text{Entropy}(S) = \sum_{\text{for all classes}} -P_{xxx} \log_2(P_{xxx})$$

$xxx$ : amphi, bird, fish, mammal, reptile

$$= -P_{amphi} \log_2(P_{amphi}) - P_{bird} \log_2(P_{bird}) - P_{fish} \log_2(P_{fish}) - P_{mammal} \log_2(P_{mammal}) - P_{reptile} \log_2(P_{reptile})$$

$$= -(3/10) \log_2(3/10) - (2/10) \log_2(2/10) - (2/10) \log_2(2/10) - (2/10) \log_2(2/10) - (1/10) \log_2(1/10)$$

$$= 2.2464$$

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Calculation of entropy for the feature give birth : $No(S_{A1}, No)$

| Name | gives birth | aquatic animal | aerial animal | has legs | Class label |
|---|---|---|---|---|---|
| Python | no | no | no | no | reptile |
| Salmon | no | yes | no | no | fish |
| frog | no | semi | no | yes | amphibian |
| pigeon | no | no | yes | yes | bird |
| turtle | no | semi | no | yes | amphibian |
| Salamander | no | semi | no | yes | amphibian |

Four class labels appear in this segment, namely, amphi, bird, fish and reptile. We have:

Number of examples with class label 'amphi' = 3

'bird' = 1

'fish' = 1

reptile = 1

Total number of examples = 6

Therefore, we have:

$$\text{Entropy}(S_{A1}, No) = \sum_{\text{for all classes 'xxx'}} -P_{xxx} \log_2 (P_{xxx})$$

$$= -P_{amphi} \log_2 (P_{amphi}) - P_{bird} \log_2 (P_{bird}) - P_{fish} \log_2 (P_{fish}) - P_{reptile} \log_2 (P_{reptile})$$

$$= -(3/6) \log_2 (3/6) - (1/6) \log_2 (1/6) - (1/6) \log_2 (1/6) - (1/6) \log_2 (1/6)$$

$$= 1.7925$$

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## Computation of Grain (S, gives birth)

$A_1$ = gives birth

Values of $A_1$ : {'yes', 'no'}

$S_{A1}$ : yes = Data in Table

$|S_{A1} \cdot yes| = 4$

Entropy $(S_{A1} = yes) = 1.5$

$S_{A_1}$ : no = Data in table

$|S_{A1} \cdot no| = 6$

Entropy $(S_{A1} = no) = 1.7925$

Now we have

$$\text{Grain}(S, A_1) = \text{Entropy}(S) - \sum_{v \in \text{Values}(A_1)} \frac{|S_v|}{|S|} \times \text{Entropy}(S_v)$$

$$= \text{Entropy}(S) - \frac{|S_{A1} \cdot yes|}{|S|} \times \text{Entropy}(S_{A1 \cdot yes}) - \frac{|S_{A1} \cdot no|}{|S|} \times \text{Entropy}(S_{A1 \cdot no})$$

$$= 2.2464 - (4/10) \times 1.5 - (6/10) \times 1.7925$$

$$= 0.5709$$

Consider the data S given in Table. Let A denote the attribute 'gives birth'. We have already seen that

$|S| = 10$

Entropy $(S) = 2.2461$

Grain $(S, A) = 0.5709$

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Now we have

$$\text{Split Information}(S, A) = -\frac{|S_{yes}|}{|S|}\log_2\frac{|S_{yes}|}{|S|} - \frac{|S_{no}|}{|S|}\log_2\frac{|S_{no}|}{|S|}$$

$$= -\frac{4}{10} \times \log_2 \frac{4}{10} - \frac{6}{10} \times \log_2 \frac{6}{10}$$

$$= 0.9710$$

$$\text{Gain Ratio} = \frac{0.5709}{0.9710} = 0.5880$$

In a similar way we can compute the gain ratios Gain(S, 'aquatic') Gain(S, 'aerial') and Gain (S, 'has legs').

# Module 3

## Concept of Classification Rules

Classification rules represent knowledge in the form of logical if else statements that assign a class to unlabeled examples.

## One Rule Algorithm (1R Algorithm)

The way this algorithm works is simple. For each feature,

Recall that the means of $x$ and $y$ are given by

$$\bar{x} = \frac{1}{n} \sum x_i$$

$$\bar{y} = \frac{1}{n} \sum y_i$$

and also that the variance of $x$ is given by

$$var(x) = \frac{1}{n-1} \sum (x_i - \bar{x_i})^2$$

The covariance of $x$ and $y$, denoted by $Cov(x,y)$ is defined as

$$Cov(x,y) = \frac{1}{n-1} \sum (x_i - \bar{x_i})^2 (y_i - \bar{y})$$

It can be shown that the values of a and b can be

$$b = \frac{Cov(x}{}$$

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Problem (Example data for simple linear

| $x$ | 1.0 | 2.0 | 3.0 | 4.0 | 5.0 |
|---|---|---|---|---|---|
| $y$ | 1.00 | 2.00 | 1.30 | 3.75 | 2.25 |

$n = 5$

$$\bar{x} = \frac{1}{5}(1.0 + 2.0 + 3.0 + 4.0 + 5.0)$$

$$= 3.0$$

$$\bar{y} = \frac{1}{5}(1.00 + 2.00 + 1.30 + 3.75 + 2.25)$$

$$= 2.06$$

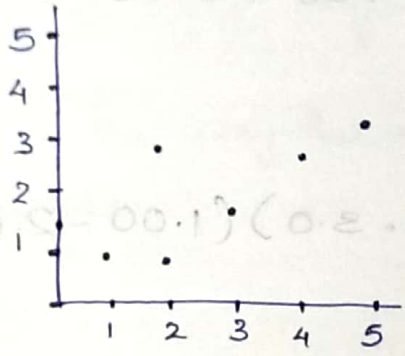
$$\text{Cov}(x, y) = \frac{1}{4}[(1.0 - 3.0)(1.00 - 2.06)$$

$$= 1.0625$$

# Problem

Use k-means clustering algorithm to divide the following data into two clusters and also compute the representative data points for the clusters.

| | x | y | | | |
|---|---|---|---|---|---|
| $x_1$ | 1 | 2 | 2 | 3 | 4 | 5 |
| $x_2$ | 1 | 1 | 3 | 2 | 3 | 5 |




Step 1: In the problem, the required numbers of clusters is 2 and we take $k = 2$.

2: We choose two points arbitrarily as the initial clusters. Let us choose arbitrarily (see)

$\bar{V}_1 = (2, 1)$, $\bar{V}_2 = (2, 3)$

3: We compute the distances of the given data points from the cluster centers. The distance between the points $(x_1, x_2)$ and $(y_1, y_2)$ will be calculated using the familiar distance formula of elementary analytical geometry:

$$\sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}$$ (Next Page)

$\sqrt{\phantom{x}}$

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| $\bar{x}_i$ | Data point | Distance from $\bar{v}_1 = (2,1)$ | Distance from $\bar{v}_2 = (2,3)$ | Minimum distance | Assigned Center |
|---|---|---|---|---|---|
| $\bar{x}_1$ | (1,1) | 1 | 2.24 | 1 | $\bar{v}_1$ ✓ |
| $\bar{x}_2$ | (2,1) | 0 | 2 ~~0+1+2~~ | 0 | $\bar{v}_1$ |
| $\bar{x}_3$ | (2,3) | 2 | 0 | 0 | $\bar{v}_2$ |
| $\bar{x}_4$ | (3,2) | 1.41 | 1.41 | 0 | $\bar{v}_1$ |
| $\bar{x}_5$ | (4,3) | 2.83 | 2 | 2 | $\bar{v}_2$ |
| $\bar{x}_6$ | (5,5) | 5 | 3.60 | 3.60 | $\bar{v}_2$ |

Cluster 1: $\{\bar{x}_1, \bar{x}_2, \bar{x}_4\}$ represented by $\bar{v}_1$

Number of data points in Cluster 1: $C_1 = 3$

Cluster 2: $\{\bar{x}_3, \bar{x}_5, \bar{x}_6\}$ represented by $\bar{v}_2$

Number of data points in Cluster 2: $C_2 = 3$.

4: The cluster are recalculated as follows:

$$\bar{v}_1 = \frac{1}{C_1}(\bar{x}_1 + \bar{x}_2$$

$$= \frac{1}{4}(\bar{x}_1 + \bar{x}_2 + \bar{x}_3 + \bar{x}_4$$

$C_1 = (1,1)(2,1)(3,2)$

$C_2 = (2,3)(4,3)(5,5)$

$$\frac{(1,1)(2,1)(3,2)}{3} = \frac{1+2+3}{3}, \frac{1+1+2}{3}$$

$= 4/3 = 1.3$

$= 6/3 = 2$

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$$\frac{(2,3)(4,3)(5,5)}{3} \cdot \frac{2+4+5}{3}, \frac{3+3+5}{3}$$

$$: \frac{11}{3} : \underline{3.2}$$

$$: \frac{11}{3} : \underline{3.2}$$

Data point dist from (2,1)

$(1,1) : \sqrt{(1-2)^2 + (1-1)^2} : 1$

$(2,1) : \sqrt{(2-2)^2 + (1-1)^2} : 0$

$(2,3) : \sqrt{(2-2)^2 + (3-1)^2} : 2$

$(3,2) : \sqrt{(3-2)^2 + (2-1)^2} : \sqrt{2}$

$(4,3) : \sqrt{(4-2)^2 + (3-1)^2} : \sqrt{8}$

$(5,5) : \sqrt{(5-2)^2 + (5-1)^2} : 5$

dist from (2,3)

$(1,1) : \sqrt{(1-2)^2 + (1-3)^2} : \sqrt{5} : 1$

$(2,1) : \sqrt{(2-2)^2 + (1-3)^2} : 2$

$(2,3) : \sqrt{(2-2)^2 + (3-3)^2} : 0$

$(3,2) : \sqrt{(3-2)^2 + (2-3)^2} : \sqrt{2}$

$(4,3) : \sqrt{(5-2)^2 + (5-3)^2} : \sqrt{13}$

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Step 4

Cluster 1 : $\{x_1, x_2, x_4\}$

$= (1,1), (2,1), (3,2)$

Cluster 2 : $\{x_3, x_5, x_6\}$

$= (2,3), (4,3), (5,5)$

$V_1 = \frac{1}{3}(x_1 + x_2 + x_4)$

$= (2, 1.33)$

$V_2 = \frac{1}{3}(x_3 + x_5 + x_6)$

$= (3.67, 3.67)$

Cluster center 1 $(2, 1.33)$

$(1,1) : \sqrt{(1-2)^2 + (1-1.33)^2} = 1.05$

$(2,1) : \sqrt{(2-2)^2 + (1-1.33)^2} = 0.33$

$(2,3) : \sqrt{(2-2)^2 + (3-1.33)^2} = 1.67$

$(3,2) : \sqrt{(3-2)^2 + (2-1.33)^2} = 1.20$

$(4,3) : \sqrt{(4-2)^2 + (3-1.33)^2} = 2.60$

$(5,5) : \sqrt{(5-2)^2 + (5-1.33)^2} = 4.74$

Cluster Center : $(3.67, 3.67)$

$(1,1) : \sqrt{(1-3.67)^2 + (1-3.67)^2} = 3.77$

$(2,1) : \sqrt{(2-3.67)^2 + (1-3.67)^2} = 3.14$

$(2,3) : \sqrt{(2-3.67)^2 + (3-3.67)^2} = 1.80$

$(3,2) : \sqrt{(3-3.67)^2 + (2-3.67)^2} = 1.80$

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$(4, 3)$: $\sqrt{(4-3.67)^2 + (3-3.67)^2} = 0.75$

$(5, 5)$: $\sqrt{(5-3.67)^2 + (5-3.67)^2} = 1.89$

| $x_i$ | Data Point | Dist from $V_1$ $(2, 1.35)$ | Dist from $V_2$ $(3.02, 3.67)$ | min | Assigned center |
|---|---|---|---|---|---|
| | | 1.05 | 3.77 | 1.05 | $V_1$ |
| | | 0.33 | 3.14 | 0.33 | $V_1$ |
| | | 1.67 | 1.80 | 1.67 | 1 |
| | | 1.20 | 1.80 | 1.20 | 1 |
| | | 2.60 | 0.75 | 0.75 | 2 |
| | | 4.14 | 1.89 | 1.89 | 2 |

# Module 5

## Evaluating Model Performance, Improving Model Performance

### Measuring Performance for classification

Matrix

| Model Predictions | Actual Results: Positive | Actual Results: Negative |
|---|---|---|
| Positive | True Positive<br>The number of observations the model predicted were positive that were actually positive | False Positive<br>The number of observations the model predicted were positive that were actually negative |
| Negative | False Negative<br>The number of observations the model predicted were negative that were actually positive. | True Negative<br>The number of observations the model predicted were negative that were actually negative |

True Positive (TP):

False Positive (FP):

True Negative (TN):

False Negative (FN):

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Accuracy = $\frac{TP + TN}{TP + TN + FP + FN}$

Confusion Matrix

| TP | FP |
|---|---|
| FN | TN |

Precision $\frac{TP}{TP + FP}$

Sensitivity $\frac{TN}{FN + TN}$

Recall $\frac{TP}{TP + FN}$ (Sensitivity)

Specificity $\frac{TN}{FP + TN}$

## Sensitivity and Specificity

The sensitivity of a model (also called the true positive rate), measures the proportion of positive examples that were correctly

The specificity of a model (also called the ~~false~~ true negative rate), measures

Problem 1

Let there be 10 balls, (6 white and 4 red balls) in a box and let it be required to pick up the red balls from them. Suppose we pick up 7 balls as the red balls of which only 2 are actually red balls. What are the values of precision and recall in picking red ball?

Solution

Obviously we have:

$$TP = 2$$
$$FP = 7 - 2 = 5$$
$$FN = 4 - 2 = 2$$

The precision P is

$$P = \frac{TP}{TP + FP} = \frac{2}{2+5} = \frac{2}{7}$$

The recall R is

$$R = \frac{TP}{TP + FN} = \frac{2}{2+2} = \frac{2}{4}$$

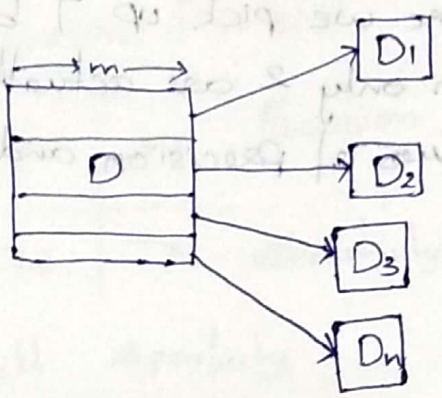
Cross - Validation

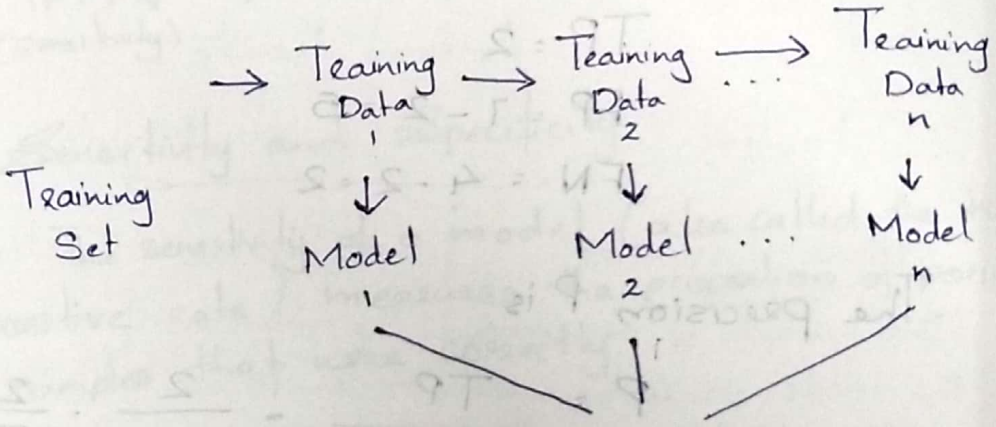
It is a technique to evaluate predictive models by partitioning the original sample into a training set to train the model, and a test set to evaluate it

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# Bootstrap Sampling

- Row Sampling with replacement

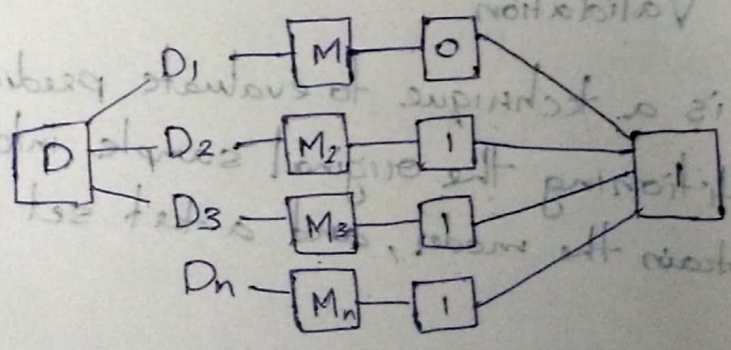



Test Set

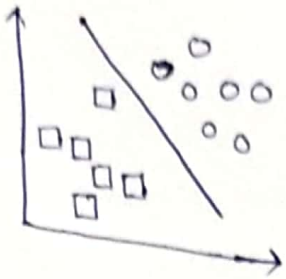
# Improving Model Performance

## 1. Ensemble Methods

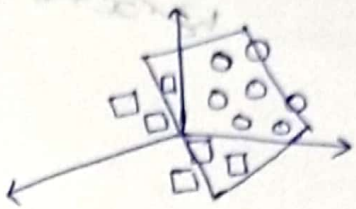



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Two Dimensions

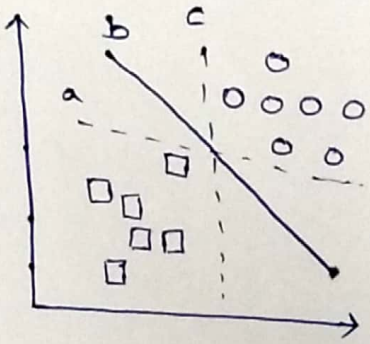


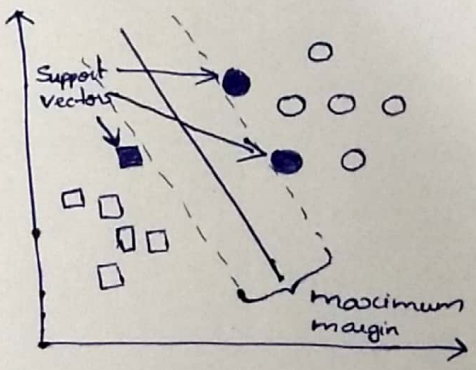
Three Dimensions



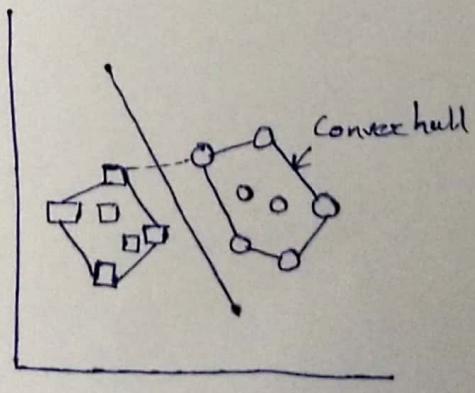
SVM - Support Vector Machine

Finding Maximum margin hyperplanes in linearly separable data




MMH - Maximum Margin Hyperplane




n-dimensional space

$\vec{w} \cdot \vec{x} + b = 0$

w - weight
b - bias
x - training data

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Set of weights that specify two hyperplanes

$$\vec{w} \cdot \vec{x} + b \geq +1$$
$$\vec{w} \cdot \vec{x} + b \leq -1$$

